Real-Time Road Damage Detection and Infrastructure Evaluation Leveraging Unmanned Aerial Vehicles and Tiny Machine Learning

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***Abstract*—Road damage detection (RDD) through computer vision and deep learning techniques can ensure the safety of vehicles and humans on the roads. Integrating unmanned aerial vehicles (UAVs) in RDD and infrastructure evaluation (IE) has also emerged as a key enabler, contributing significantly to data acquisition and real-time monitoring of road damages, such as potholes, cracks, and surface anomalies, facilitating proactive maintenance and improved road conditions. These UAVs are low-powered and resource-constrained devices that work autonomously to perform pattern detection and decision making leveraging tiny machine learning (Tiny ML) algorithms. These Tiny ML algorithms are designed to run on edge devices, IoT devices, UAVs, etc. In this study, the RDD2022 data set collected using UAVs and dashboard cameras of vehicles was utilized to train pure and mixed models that exhibit class instance imbalance in certain classes which is addressed by implementing data augmentation as a regularization technique. State-of-the-art two-stage detectors: Faster R-CNN ResNet101 and one-stage detectors: SSD MobileNet V1 FPN, YOLOv5, and Efficientdet D1 are employed. The results indicate that the two-stage detector achieved an impressive mean average precision (mAP) of 88.49% overall and 96.62% for focused classes. Notably, the state-of-the-art Efficientdet D1 approach achieved a competitive mAP of 86.47% overall and 95.12% for focused classes, with significantly lower computational cost. These findings highlight the potential of advanced object detection techniques, particularly Efficientdet D1, to enhance the accuracy and efficiency of RDD systems, thereby improving passenger safety and overall performance.**

***Index Terms*—Efficientdet, faster R-CNN, object detection (OD), one-stage detectors, road damage detection (RDD), SSD mobileNet, tiny machine learning (Tiny ML), two-stage detectors, unmanned aerial vehicles (UAVs), YOLOv5.**

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I. Introduction

In RECENT years, the rapid advancements in artificial intelligence (AI) technologies have paved the way for transformative developments across various smart applications, such as unmanned aerial vehicle (UAVs) and tiny machine learning (Tiny ML) [1]. The Tiny ML is a type of machine learning (ML) algorithm that is designed for low-power, resource-constrained devices, enhancing automation, and decreasing the workload on humans [2]. These devices are edge devices, IoT devices, UAVs, etc. UAVs also known as drones are remotely controlled or autonomous devices with the capability to fly without any human operator [3]. The TinyML enables UAVs for data collection, data analysis, and detecting patterns for decision making. The amalgamation of Tiny ML with UAVs represents a transformative leap forward in the domain of road damage detection (RDD) to ensure the safety of both humans and vehicles [4].

Several factors contribute to road damage (RD) like heavy traffic, weather conditions, and natural disasters [5]. The most common RDs include different types of cracks, bumps, and potholes [6]. These RDs can cause accidents and other safety issues, as well as increase maintenance costs for road authorities [7]. Therefore, it is critical to detect, repair, and maintain RDs to avoid an uncomfortable situation. The early and timely detection of these RDs can also help authorities repair them on time.

Discussed RD conditions normally can be detected using three methods: 1) manual; 2) semi-automated; and 3) fully automated. The manual method is a subjective judgment that involves observing the road surface by either walking

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or using a slow-moving vehicle. It is time consuming and requires significant human intervention [8]. Moreover, in a semi-automated approach, images are collected with a camera mounted on a fast-moving vehicle; however, human effort is still required for damage identification [9]. In a fully automated approach, the images are taken using a UAV or the camera mounted on a vehicle [10] and are fed into pattern recognition software to identify RDs in real time. There has been significant progress in the development of object detection (OD)-based automated RDD and infrastructure evaluation (IE) systems. The data to train such a system can be acquired through UAVs which enables comprehensive surveys of road conditions, such as potholes, cracks, signage, lane markings, and infrastructure readiness, leveraging their aerial perspective and facilitating swift and efficient data acquisition.

One of the initial methods for RDD and IE was based on traditional image processing (IP) techniques. These techniques involved extracting features from images and using them to classify RDs [11]. The typical approach in classical IP for OD is the use of thresholding to distinguish the object from its background [12]. Another similar study [13] employed grayscale histogram and Otsu thresholding to identify pavement cracks by dividing the input image into subimages based on the maximum histogram to threshold value ratio. Similarly, the Otsu thresholding algorithm and gray level co-occurrence matrices (GLCMs) were also used by [14] to extract road crack features, which were then classified using the support vector machine (SVM) algorithm. These methods yielded reasonable accuracy but may face challenges in generalizing to new and diverse road conditions due to their reliance on manual feature engineering.

To overcome the limitations of traditional IP techniques for RDs and IE, DL techniques can be utilized. Deep neural networks (DNNs) are capable of learning complex patterns and backgrounds by extracting low-, middle-, and high-level features from image data and as a result, they can detect complex RDs [15]. DL-based approaches for RDD and IE usually include three processing steps: 1) data collection; 2) data annotation; and 3) training [16]. Various DL techniques, such as SSD [17], Faster R-CNN [18], [19], YOLOV7 [20], [21], EfficientNet [22], and EfficientDet [23] have been used for RDD and IE [24], [25], [26]. Reference [27] also used DL techniques to detect multiple types of RD, such as potholes, alligator cracks, etc.

Shaghouri et al. [28] compared the effectiveness of SSD TensorFlow, YOLOv3-Darknet53, and YOLOv4-CSPDarknet53 in detecting potholes, with YOLOv4 exhibiting the best results. YOLOv4 was able to detect potholes at 100 m, with a speed of 20 frames/s, achieving a high recall rate of 81%, a high precision rate of 85%, and a mean average precision (mAP) of 85.39%. Another study [29] proposed a method that employed Faster-RCNN as the detection framework and ResNet-152 as the feature extraction network to detect and classify RDs. However, the large number of network parameters resulted in slow processing speeds and limited multiscale detection capability. Yebes et al. [30] also used Faster-RCNN with Resnet101 as the feature extraction network to achieve faster processing speeds, but the accuracy was only 75% even with the Intersection over Union (IoU) index relaxed to 0.4 for evaluation, and the speed was only 5–6 frames/s. A pothole localization method using thermal images was presented by [31] based on the SSD and RetinaNet detection frameworks with ResNet34 and ResNet50 as the feature extraction networks, to address detection challenges caused by unfavorable weather conditions with low visibility.

In a recent study, an enhanced and lightweight YOLOv5 (YOLO-LRDD) approach was proposed for RDD and IE. This approach addressed the issues of complex models and time-consuming computations in current DL-based RDD methods. Compared to YOLOv5, YOLO-LRDD achieved a 22.3% improvement in single image recognition speed and a 28.8% reduction in model size, while still maintaining similar accuracy. Hence, it achieves a satisfactory trade-off between detection accuracy and processing speed [32]. Another study [33] proposed a DL model to identify the RDs and IE in real time. To enhance the generalizability of the trained models for deployment in countries not included in the training data, they employed two approaches: 1) pure models, which were trained solely on data sets from a single country and 2) mixed models, which were trained on data sets from multiple countries. This work reported a better mAP (%) of about 92% for the focused class by using a single-stage OD model. These techniques have been found to improve detection accuracy, as well as to estimate the severity of damages on the road [34]. This study aims to develop an advanced RDD system that can efficiently monitor and detect RDs in real time. For this purpose, the RDD2022 data set [6] from three countries, including the United States, China, and Japan, was used to deploy the models. This data set includes images taken by vehicle-mounted cameras and UAVs consisting of five different classes named D00, D10, D20, D40, and repair. The collected data set was imbalanced where some classes were significantly underrepresented, leading to misclassification and inadequate representation of important minority classes. Balancing the representation of instances across all classes improves model accuracy, performance, and decision making. To overcome this problem, data augmentation was performed which improved the performance of mixed models significantly. Four types of DL models, including Faster R-CNN, YOLOv5, SSD MobileNet, and EfficientDet, were trained and evaluated using this data set as pure and mixed models. This work will contribute to automating the RDD and can also help schedule timely repair and maintenance of roads to avoid accidents.

A. Motivation and Contributions

Class imbalance is a common challenge in many real-world data sets, where certain classes have significantly lower representation compared to others which can lead to misclassification and inadequate representation of important minority classes. Striving for a balanced representation of instances across all classes, can not only enhance the accuracy and performance of the model but also facilitate more effective decision making, particularly in scenarios where instances from the minority classes like D40 and repair are critical and

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require special attention. In the context of instances, such as D00, D10, D20, D40, and repair, in RDD2022, it becomes crucial to address this class imbalance to ensure accurate and reliable predictions. To address this issue data augmentation, a regularization technique was applied to ensure that the model provides reliable insights and predictions for all instances, including D00, D10, D20, D40, and repair.

Overall, the contributions of this study are given below.

1) *Data Augmentation:* A regularization technique was applied to deal with the class imbalance issue for a large data set RDD2022.
2) Development of two-stage and one-stage DL OD models for identification of RDs and IE.
3) Performance analysis and comparison of mAP values of models for focused classes after dealing with the class imbalance issue to enhance the model's capability.

These contributions aim to improve the accuracy and efficiency of RDD systems, which are important for infrastructure operations and maintenance. The research adds to the growing body of knowledge on the use of DNNs for RDD and IE and has the potential to enhance road safety for all users.

This article is organized as follows. Section I reviews the importance RDDs an IE. Section II explains the methodology used and the connectivity of physical and digital infrastructure. Section III presents the results and discussion, focusing on the potential benefits of DL models for monitoring and evaluating the digital infrastructure. Section V concludes this article and shows the need for ongoing research and development in this field.

II. Methodology

This section of study encompasses several key aspects, including data set description, data preprocessing, data augmentation, experimental setup, model setup, and evaluation criteria of the proposed model. First, the data set will be thoroughly described, highlighting its sources, composition, and relevant features. Subsequently, the data preprocessing techniques employed to ensure data quality and consistency will be outlined. Additionally, the implemented data augmentation methods to enhance the diversity and generalizability of the data set will be explained. Moreover, the experimental setup, encompassing hardware and software configurations, will be presented in detail, along with any necessary specifications and evaluation criteria to compare mAP values of state-of-the-art methods for OD. Finally, the model setup, involving the architecture, hyperparameters, and optimization algorithms, will be thoroughly explained.

A. Data Set Description

The RDD2022 is a comprehensive data set that contains about 47 000 images from six different countries, namely, China, Japan, the United States, Norway, India, and the Czech Republic. The images are taken via UAVs and dashboard cameras of the vehicles. The data set is organized into eight different classes, including D00, D10, D20, D40, D43, D44, D50, and Repair class [6]. The data set is primarily designed for studying and analyzing RDD and their classification. It contains images of various types of RDs, including potholes, cracks, and rutting, as well as images of road repairs. This study focuses on a subset of the RDD2022 data set, containing images from three different countries, namely, the United States, China, and Japan. The subset consists of five different classes, including D00, D10, D20, D40, and Repair. A detailed description of these classes is given in Table I and sample images in Fig. 1. Overall, the RDD2022 data set and its subsets provide a great resource for researchers and practitioners in the field of RDD and IE. This data set contains a diverse range of road and weather conditions, as well as a large number of images. Therefore, the data set offers a unique opportunity to develop and test new algorithms and techniques for RDD and classification.

TABLE I
Types of RDs

Damage Type	Detail	Class Name
Longitudinal Cracks	Wheel-marked part	D00
Transverse Cracks	Equal Interval	D10
Alligator Cracks	Partial pavement, Overall pavement	D20
Potholes	Pothole	D40
Repair Section	Repaired Part	Repair

TABLE II
Data Set Distribution Used for the Experimental Work

Dataset	Images	Labels	Resolution	Train_set	Val_set	Test_set
China	2000	10780	640*640	1600	200	200
Japan	2002	10901	640*640	1601	200	201
US	2000	9585	640*640	1600	200	200
Mix	3000	13553	640*640	2400	300	300
Total	**8476**	**42819**	**-**	**6781**	**797**	**850**



Fig. 1. Annotated sample images from the data set.

B. Data Preprocessing

Data preprocessing plays a crucial role in the field of ML, as it encompasses essential tasks, such as data cleaning, transformation, and preparation to ensure the raw data is ready for analysis. In this study, the RDD2022 data set is employed focusing on three specific countries, each country subset consists of 2000 carefully selected images, resulting in three distinct sets. Furthermore, a fourth composite set is created, comprising mixed images from all three countries. The detail of these four sets of images is shown in Table II. The selected images underwent a preprocessing step where they were cleaned, annotated, and prepared to be fed into detection models. The cleaning process involved removing any irrelevant or unwanted information from images, while annotation involved labeling each image with the appropriate class type. Subsequently, the prepared images were employed to train and evaluate different detection models.

C. Data Augmentation

Data augmentation is a regularization method to extend the data set artificially, preventing the overfitting issues. The new data is created by generating modified versions of the original

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TABLE III
INSTANCE COUNT OF CLASSES BEFORE AND AFTER DATA AUGMENTATION

		Instance count of classes before augmentation in %					Instance count of classes after augmentation in %				
Class Name		D00	D10	D20	D40	Repair	D00	D10	D20	D40	Repair
Dataset	China	57	24	14	5	4	33	22	17	15	13
	US	61	31	7	1	1	28	22	20	15	15
	Japan	62	30	7	1	1	30	25	20	14	11
	Mix	60	28	10	2	1	32	19	20	15	14

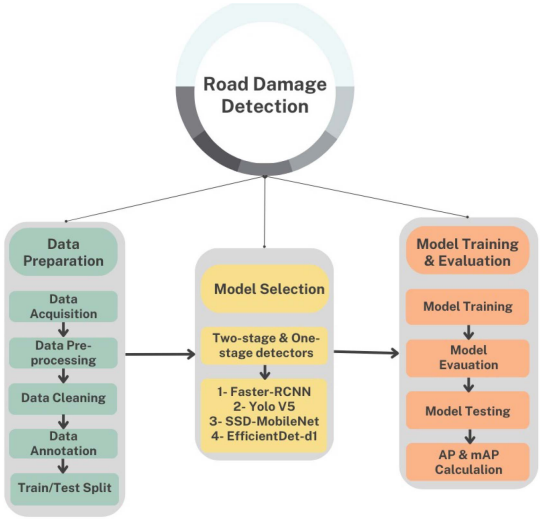


Fig. 2. Flowchart of activities carried out in this study.



Fig. 3. Sample images fed to the model for training purpose.

data. The subset of RDD2022 used for this study contained varying numbers of instances of different classes like D00 60%, D10 28%, D20 10%, D40 2%, and Repair less than 1% in mixed category data set. These percentages show that the class instances are imbalanced in the available data set that can have a detrimental effect on the model's performance, as the model may become biased toward the classes with more instances and fail to detect the less-represented classes.

To address this issue, data augmentation was used in this study as a regularization technique. Augmentation includes the application of diverse transformations, such as random cropping, horizontal and vertical flipping, rotating, scaling, brightness, and contrast adjustments. This process generates new images that represent variations of the original images to

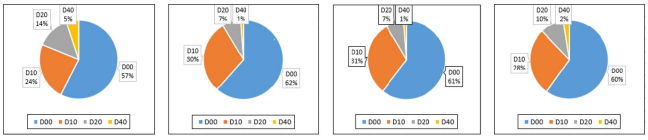


Fig. 4. Instance count of classes in (%) form before data augmentation.

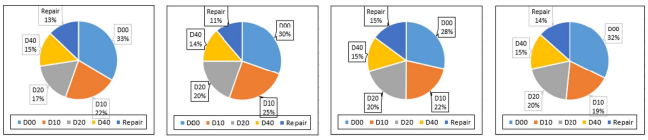


Fig. 5. Instance count of classes in (%) form after data augmentation.

increase the variability of the data set, making it less prone to overfitting. These transformations were applied randomly to the images creating a more diverse and balanced data set for the model training. The applied regularization technique led to significant improvements in class instances across the mixed-category data set. Specifically, the instances for D20 increased up to 20%, D40 by 15%, and Repair by 14%. A detailed description of classes, before and after data augmentation is given in Table III, and graphical representation is shown in Figs. 2 and 3. Through the implementation of augmentation to tackle the class imbalance issue, the impact of the imbalance was reduced, resulting in an improvement in the mAP values of the trained models.

D. Experimental Setup

The experimental setup used in this work is the Google Collaboratory Pro-Plus version. The setup includes a GPU, specifically the NVIDIA A100-SXM, which offers 40 GB of memory, making it well suited for training DL models. Additionally, the system features a large amount of RAM, 54.8 GB, enabling efficient processing of large data sets. The eight system CPUs of Intel Xeon @ 2.20 GHz provide the computational power to support data processing tasks and allow for rapid model training.

E. Model Setup

In DL, OD models typically consist of multiple convolutional layers followed by feature extraction and prediction layers. The specific architecture may vary based on the model type, such as one-stage or two-stage detectors. The overall flow of the system activities consisting of data preparation, model selection, model training, and evaluation is shown in Fig. 4. It presents the sequence of steps followed to perform experiments using different models. This study implements the

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TABLE IV
HYPERPARAMETERS OF OD MODELS FINE-TUNED DURING EXPERIMENTS

Model_Name	Hyper_parameters									
	Input_size	Batch_size	Steps	Epochs	Optimizer	Reg_tech	Learning_rate			
Faster_rcnn_resNet101	640*640	8	35000	1	Momentum	L2	US	China	Japan	Mix
YOLOv5	640*640	8	35000	1	Momentum	L2	0.04	0.04	0.04	0.04
SSD_mobileNet_v1-fpn	640*640	8	35000	1	Momentum	L2	0.04	0.04	0.04	0.04
Efficientdet-d1	640*640	8	35000	1	Momentum	L2	0.04	0.04	0.04	0.04

training of various models utilizing cutting-edge DL architectures. The training can be categorized into two approaches.

1) *Pure Models:* These models employ image data collected from a single country/source.
2) *Mixed Models:* These models combine data sets sourced from various countries to enhance the models' generalization capabilities.

Subsequently, these models are tested using data obtained from the target country, as well as images from different countries.

1) Faster R-CNN ResNet101: The Faster R-CNN ResNet101 stands as a two-stage OD model that has consistently attained state-of-the-art results in numerous benchmark data sets, including COCO [33], VOC [35], and KITTI [17], demonstrating its effectiveness in OD tasks. Due to its remarkable accuracy and efficiency, this model has gained popularity in various applications, such as autonomous driving, robotics, and surveillance systems [18].

The model is based on a region proposal network (RPN) and a Fast R-CNN detection network. In the initial stage, the RPN takes an input image and generates a set of candidate object proposals. It achieves this through a fully convolutional network (FCN) that predicts the objectness score and the coordinates of the proposals for each position on the feature map. These proposals are then passed on to the second stage of the model. In the second stage, the Fast R-CNN detection network refines the proposals generated by the RPN and conducts object classification and localization. This network incorporates a Region of Interest (RoI) pooling layer to extract features from the proposed regions, followed by a series of fully connected layers responsible for classification and regression tasks [36]. Overall model specifications are given in Table IV.

F. YOLOv5

YOLOv5 is an efficient one-stage object detector that utilizes a deep CNN for accurate OD in images. This model architecture processes the entire image at once and makes real-time OD possible. The methodology of YOLOv5 involves the following steps. First, collect and annotate a data set of images with objects to be detected. Second, fine-tune the pretrained YOLOv5 model for the annotated data set using a GPU. Third, the model is trained using a combination of classification and regression losses to optimize for both OD accuracy and localization accuracy [37]. Thereafter, evaluate the performance of the trained model on a validation set, and if required, perform fine-tuning. The evaluation of the model's performance involves using metrics like mAP and IoU. Finally, deploy the trained model for OD on new images or videos. The YOLOv5 model specifications are also shown in Table IV.

1) SSD MobileNet V1 FPN: The SSD MobileNet v1 FPN is a popular OD model; built upon a full CNN that executes OD in a single pass. Comprising a MobileNet v1 architecture as the backbone network and an FPN, it generates feature maps at multiple scales. The backbone network extracts features from the input image, while the FPN combines these features to produce a set of feature maps utilized for generating object predictions. The model employs default boxes, and predefined anchor boxes at various aspect ratios and scales to generate object proposals. These proposals are refined using regression, and the model performs object classification using a softmax function [33], [36]. The model specifications are also given in Table IV.

2) Efficientdet-d1 Model: The EfficientDet D1 model represents a powerful and efficient OD model that demonstrates the effectiveness of efficient architecture design choices. Through a novel combination of efficient design principles, the model achieves remarkable accuracy and efficiency. Its capability to deliver state-of-the-art results while requiring fewer computational resources, it is a promising and practical choice for real-world OD applications [38]. The model specification details are also given in Table IV. The EfficientDet D1 model architecture is based on the EfficientNet backbone network, which is optimized for both accuracy and efficiency using a compound scaling method. In compound scaling, the depth, width, and resolution of the model architecture increase in a constant manner, as shown in the following equations:

$$f = \alpha \cdot \beta^{\varphi} \cdot \gamma^{\varphi} \tag{1}$$

$$f = d \cdot \beta^{\varphi} \cdot \gamma^{\varphi} \tag{2}$$

where α is "d" which is depth scaling factor, β is "w" as a width scaling factor, γ is "r" known as a resolution scaling factor, and "f" is a network scaling factor.

EfficientNet has been successful in image classification tasks by achieving high accuracy while optimizing for efficiency parameters, such as model size and floating-point operations (FLOPS). EfficientDet builds on this success by applying similar principles to OD. The model is available in a range of sizes from d0 to d7, with each size being optimized for a different level of computational resources. In addition, the model incorporates a bidirectional feature pyramid network (BiFPN) module. That enhances the efficiency of the model by effectively integrating multiscale features from different layers. BiFPN enables bidirectional information flow, balancing fine-grained details and contextual information. This makes the model well suited for various computer vision tasks, where achieving accuracy while managing computational resources is crucial [23], [38].

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TABLE V
AVERAGE PRECISION AND MAP OF MODELS IN (%)

Dataset	Model Name	AP (%)					mAP(%)
		D00	D10	D20	D40	Repair	
China	SSD_mobileNet_v1-fpn	34.72	37.09	69.06	92.21	92.86	65.18
	YOLOv5	39.32	48.82	79.09	90.11	94.43	70.354
	Efficientdet-d1	32.72	36.09	63.06	94.61	95.01	64.29
	Faster_rcnn_resNet101	41.72	51.09	81.06	96.38	96.77	73.40
US	SSD_mobileNet_v1-fpn	23.09	15.29	33.74	89.60	90	50.33
	YOLOv5	27.31	21.35	38.51	90.2	91.23	53.72
	Efficientdet-d1	10.71	11.33	21.41	91.35	92.17	45.39
	Faster_rcnn_resNet101	30.10	31.26	43.11	96.55	96.73	59.55
Japan	SSD_mobileNet_v1-fpn	20.97	13.78	26.37	90.64	91.03	48.55
	YOLOv5	21.51	17.01	28.30	92.59	93.21	50.52
	Efficientdet-d1	17.41	18.65	31.54	95.88	95.83	51.86
	Faster_rcnn_resNet101	39.10	35.91	42.70	96.75	96.91	62.26
Mix	SSD_mobileNet_v1-fpn	70.61	71.09	79.16	91.44	94.49	81.35
	YOLOv5	74.98	69.13	87.15	95.91	95.21	84.41
	Efficientdet-d1	70.76	77.82	87.23	96.13	96.43	86.47
	Faster_rcnn_resNet101	81.01	79.81	87.23	97.63	96.81	88.49

Additionally, the model implements an anchor-based OD, by creating a set of anchor boxes with varying scales and aspect ratios, while predicting the offset and confidence for each anchor box. To handle the issue of class imbalance, the model employs a focal loss function. Moreover, the model stands out due to its remarkable efficiency. Despite its impressive performance, it utilizes notably fewer parameters and FLOPs compared to competing models. This makes it a suitable choice for utilization on edge devices with constrained computational resources.

III. EXPERIMENTAL RESULTS AND ANALYSIS

This section contains the experimental results of four OD models, namely, Faster R-CNN, YOLOv5, SSD MobileNet V1, and Efficientdet D1 for detecting and monitoring RDs. The four sub-data sets were used to compute the statistical values of average precision (AP) (%) and mAP (%) of the five classes: D00, D10, D30, D40, and Repair using all the models. It was observed that AP values for D40 and Repair classes were very low due to less number of instances. To improve the results data augmentation was performed and AP and mAP values were again computed for all models. The performance of these models only after data augmentation is explained below and values are shown in Table V. The section concludes by highlighting the best performing model based on mAP values.

A. Train/Test Process and Evaluation Criteria

1) Training Process: The training process of OD models for RDD and IE involved several steps. First, data preprocessing was performed to feed the models with prepared data. Second, four OD models were chosen and implemented with hyperparameter tuning, to best fit the research problem. Third, these models were trained using images from all data sets, and the training loss was calculated. Some of the sample images are shown in Fig. 5. Fourth, the backpropagation was then used to minimize the overall cost function of the model by updating the weights and biases using Adam optimizer. During the training process, the model updates the weights and biases every time using the until the training loss reaches at an acceptable level. Finally, the L2 regularization was used to prevent the model from overfitting. The training process is computationally extensive and requires specialized and powerful hardware. Therefore, in this work, GPUs were used to train the models.

2) Testing Process: The testing process is a very important step in model development that involves the evaluation of a trained model on a data set that was not used in the training phase. This process helps to check the generalization of the trained model and establish that it can also perform well on unknown data. This process involves various steps. First, the input data is divided into two subsets, namely, train and test. In this work, data was split according to a 90:10 ratio. Second, The trained model is provided with the test data to make predictions. Finally, the results are evaluated using mAP values, which is considered the state-of-the-art evaluation metric for OD.

3) Evaluation Criteria: Several statistical metrics were used to evaluate the trained OD models in real time for UAVs. The description of these metrics is given below.

1) *IoU:* It is a widely employed evaluation metric for OD models. It quantifies the agreement between predicted and ground-truth bounding boxes by computing the ratio of their intersection area to the union area. IoU values span from 0 to 1, signifying a lack of overlap (0) to a perfect match (1) between the boxes. IoU is commonly used as a threshold to distinguish true positives (TPs) from false positives (FPs) in OD tasks. A prediction is labeled as a TP if its IoU with the ground-truth bounding box exceeds a predetermined threshold (e.g., 0.5); otherwise, it is classified as an FP. The IoU of the

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bounding boxes can be calculated using

$$\text{IoU} = \frac{A \cap B}{A \cup B} \tag{3}$$

where A and B are the predicted and ground-truth bounding boxes, respectively.

2) *Precision–Recall Curve (PRC):* It holds significant importance as an evaluation metric in OD. It visualizes the relationship between precision and recall by adjusting the threshold for IoU. Precision, denoted as the ratio of TP detections to the sum of TP and FP detections, and recall, defined as the ratio of TP detections to the sum of TP and false negative (FN) detections. The precision and recall are represented by (4) and (5), respectively

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{4}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}. \tag{5}$$

The PRC serves as a valuable tool for analyzing the precision–recall tradeoff and determining the optimal threshold that achieves a balanced performance between the two metrics.

3) *AP and mAP:* These are commonly used evaluation metrics in OD and information retrieval tasks. AP measures the precision at various recall levels and calculates the average across all recall levels. It provides a comprehensive assessment of the model's precision–recall tradeoff. On the other hand, mAP calculates the mean of AP across multiple classes, providing a single performance measure for the entire data set. mAP is particularly useful when dealing with multiple object classes or retrieval queries. These metrics enable researchers to evaluate and compare the performance of OD systems accurately.

B. Model Results

1) Faster R-CNN ResNet101: In this study, the Faster R-CNN ResNet101 model was employed and its performance was assessed on the RDD2022 data set. Initially, the data set was divided into training and testing sets and the AP values for all the classes were calculated. Additionally, the mAP values for each of the four sub-data sets including China, United States, Japan, and Mix category, were computed as 73.40%, 59.55%, 62.26%, and 88.49%, respectively. The statistical values for all the four sub-data sets is shown in Table V.

Furthermore, a few detected samples of different classes, along with their corresponding AP values, are presented in Figs. 6 and 7. The results indicate that the model achieved the highest mAP value of 88.49% for the mixed data set. It is also observed that augmenting the D40 and Repair classes instances had an overall effect on the model results.

Overall, the results demonstrate the effectiveness of the Faster R-CNN ResNet101 model for OD in the context of RDD. The high mAP value obtained for the mixed data set suggests that the model can accurately detect RD of various types and severity levels, which is a crucial factor in ensuring road safety.

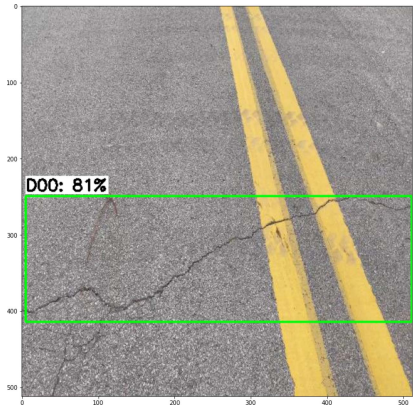


Fig. 6. Longitudinal cracks (D00) with AP value detected by trained faster_rcnn model.



Fig. 7. Alligator cracks (D20) with AP value detected by trained faster_rcnn model.

2) YOLOv5: The YOLOv5 model was utilized and assessed in this study to detect objects in the RDD data set. The data set was divided into training and testing sets, and the AP (%) values were calculated for each class. Statistical values for the four sub-data sets used in the analysis, namely, China, U.S., Japan, and Mix category, are presented in Table V. The table includes the AP values for all five classes, as well as the computed mAP values for each sub-data set: 70.35%, 53.72%, 50.52%, and 84.41%, respectively.

Several samples from different classes, along with their corresponding AP values, can be seen in Figs. 8–10. The findings show that the model attained the greatest mAP value of 84.41% for the mixed data set. Furthermore, it was found that increasing the number of instances of the D40 and Repair classes had a positive impact on the model's outcomes.

In general, the results show that the YOLOv5 model is effective for OD in the setting of RDD. The high mAP value obtained for the mixed data set demonstrates that the model can detect diverse types and severity levels of RD, which is critical for assuring road safety. On the other hand, the mAP values for specific countries were rather low, indicating that the model's effectiveness may vary based on location and type of RD. More research is required to develop robust and accurate models for specific countries.

3) SSD MobileNet V1 FPN: The SSD MobileNet V1 FPN model was used in this study to recognize objects, and its

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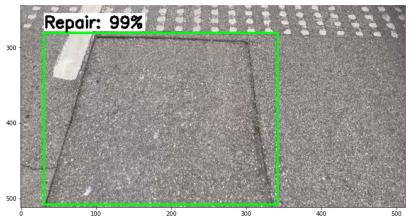


Fig. 8. Repair section with AP value detected by trained YOLOv5 model.



Fig. 9. Transverse cracks (D10) with AP value detected by trained YOLOv5 model.

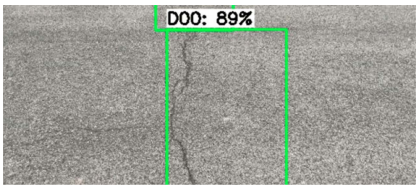


Fig. 10. Longitudinal cracks (D00) with AP value detected by trained YOLOv5 model.

performance was evaluated on the RDD2022 data set. The data set was initially separated into training and testing sets, and the AP(%) and mAP(%) values for all classes were calculated using the aforementioned model. Table V shows the statistical values for the four sub-data sets used in the investigation, namely, China, United States, Japan, and the Mix category. The model achieved mAP values of 65.18%, 50.30%, 48.55%, and 81.35%, respectively, for all sub-data sets. Figs. 11–13 show a few examples from different classes along with their corresponding AP values. The results highlight that the SSD MobileNet V1 FPN model obtained the highest mAP value of 81.35% for the mixed category data set. Furthermore, it was observed that increasing the instances of the D40 and Repair classes had significantly improved the model's performance.

The results demonstrate the effectiveness of the SSD MobileNet V1 FPN model for OD in the context of RDD. The high mAP value obtained for the mixed category data set suggests that the model can accurately detect RD of various types and severity levels, which is critical for ensuring road maintenance and passenger safety.

4) EfficientDet D1: In this work, the Efficientdet D1 model was applied for OD and evaluated its performance on the RDD2022 data set. First, the data set was divided into training



Fig. 11. Pothole (D40) with AP value detected by trained ssd mobileNet model.



Fig. 12. Longitudinal cracks (D00) with AP value detected by trained ssd mobileNet model.



Fig. 13. Alligator cracks (D20) with AP value detected by trained ssd mobileNet model.

and testing sets, and then the AP (%) and mAP(%) values for all the sub-data sets, including China, United States, Japan, and Mix category, were computed. These statistical values for all four sub-data sets are shown in Table V. The mAP values for each of the four sub-data sets are 64.29%, 45.39%, 51.86%, and 88.49%, respectively.

Furthermore, a few detected samples of different classes along with their corresponding AP values, are shown in Figs. 14–16. Moreover, the different types of losses, learning rate, and steps per second during the training phase are represented in Fig. 17. The results indicate that the model achieved the highest mAP value of 86.47% for the mixed data set. It was also observed that augmenting the instances of D40 and Repair classes improved the computed results

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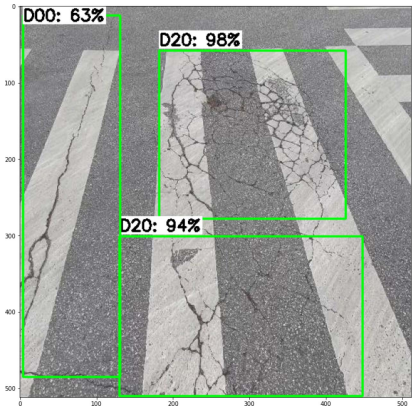


Fig. 14. Longitudinal (D20) and alligator (D00) cracks with AP values detected by trained efficientdet-d1 model.

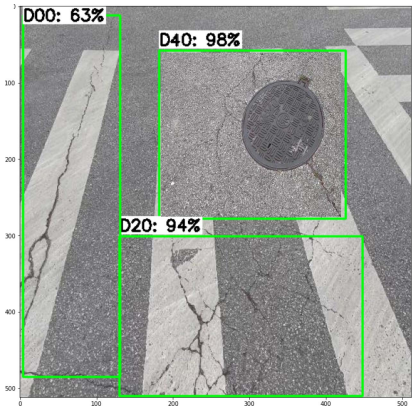


Fig. 15. Transverse (D10), longitudinal (D20) cracks, and pothole (D40) with AP values detected by trained efficientdet-d1 model.

significantly. The results demonstrate the effectiveness of the Efficientdet D1 model for OD in the context of RDD. The high mAP value obtained for the mixed data set suggests that the model can accurately detect RD of various types and severity levels, which is a crucial factor in ensuring road safety, maintenance, and IE. The Efficientdet d1 model, using the innovative BiFPN and compound scaling, emerged as the superior performer among all the one-stage models employed in this study. This advanced architecture not only achieved outstanding performance, rivaling that of two-stage models but also with a reduced parameter count, making it a compelling choice for scenarios where a tradeoff between accuracy and available computational resources is required. The incorporation of the BiFPN network played a pivotal role in achieving this high performance by enhancing the fusion of features across different levels of the feature pyramid, facilitating improved context and detail integration, and ultimately contributing to the model's impressive efficiency and effectiveness in RDD.

IV. Discussion

All the OD models implemented in this study, including Faster R-CNN ResNet101, YOLOv5, SSD MobileNet V1 FPN, and Efficientdet D1, performed well on the RDD2022 data set for RDD and IE via UAVs and dashboard cameras.

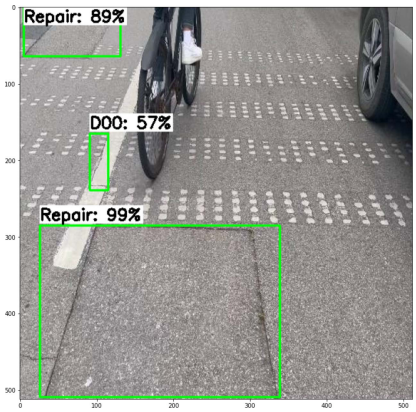


Fig. 16. Repair section with AP values detected by trained efficientdet-d1 model.

The models were evaluated based on their AP (%) and mAP (%) values. The two-stage object detector Faster R-CNN ResNet101 trained as a mixed model achieved the highest mAP value of 88.49%. The highest performance of the Faster R-CNN with ResNet101 as the backbone, trained as a mixed model, can be attributed to several factors. First, the ResNet101 architecture boasts deep residual layers that capture intricate patterns, ensuring discriminative feature extraction, which is crucial for OD. Second, the Faster R-CNN, with its RPN, efficiently proposes candidate object locations, further fine-tuning object localization. Collectively, the depth of ResNet101 and the effective training strategy ensure optimal precision and recall tradeoffs. Moreover, among the one-stage detectors, the state-of-the-art Efficientdet D1 achieved the highest mAP value of 86.47% on the Mix category data set closely followed by YOLOv5 and SSD MobileNet, which achieved an mAP value of 84.41% and 81.35%, respectively, as shown in Table V.

These experimental results suggest that all four models are capable of accurately detecting RDs, while the Efficientdet D1 model achieves promising results being more efficient and faster than the other models. This model achieves results nearly close to a two-stage detector Faster R-CNN but with comparatively less cost. This is due to the use of BiFPN which optimizes feature fusion and information flow within the network, reducing the need for computationally expensive stages like RPNs and post-processing. This efficiency allows it to keep a balance between accuracy and computational cost, making it an attractive choice for OD tasks where resource efficiency is a priority without compromising much on performance. It is also worth noting that Efficientdet-d1 model achieved the highest mAP value 95.12% against the focused classes as shown in Table VI and graphical explanation is shown in Fig. 18.

The individual AP (%) values for each class indicated that the models performed better in detecting D00, D10, and D20 classes, while the models struggled with the D40 and Repair classes. This could be due to the low number of instances for these classes in the RDD2022 data set, which might have caused the models to have limited exposure and learning on how to accurately detect the D40 and Repair class. In

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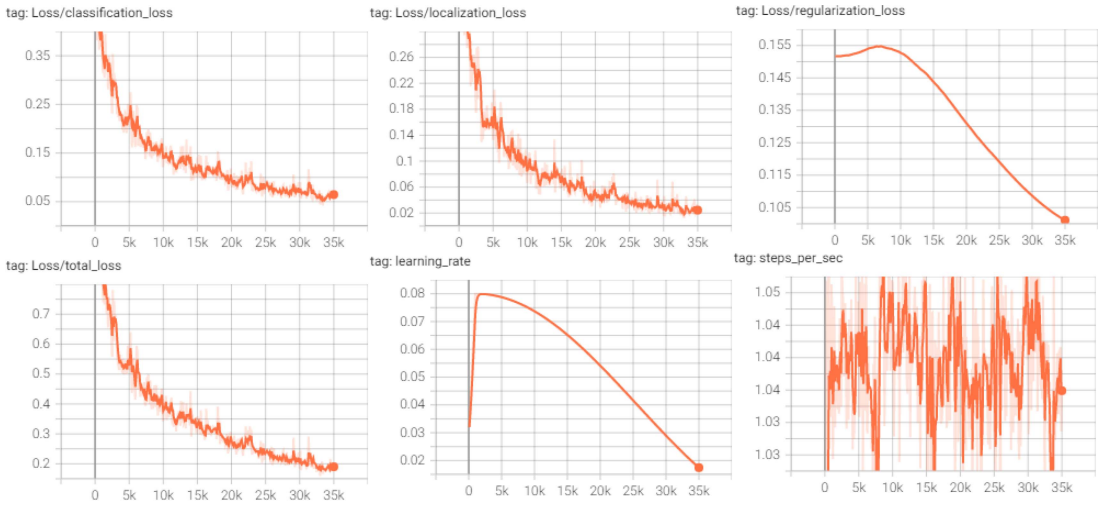


Fig. 17. Classification, localization, regularization, total loss, learning rate, and steps per second of efficientdet-d1.

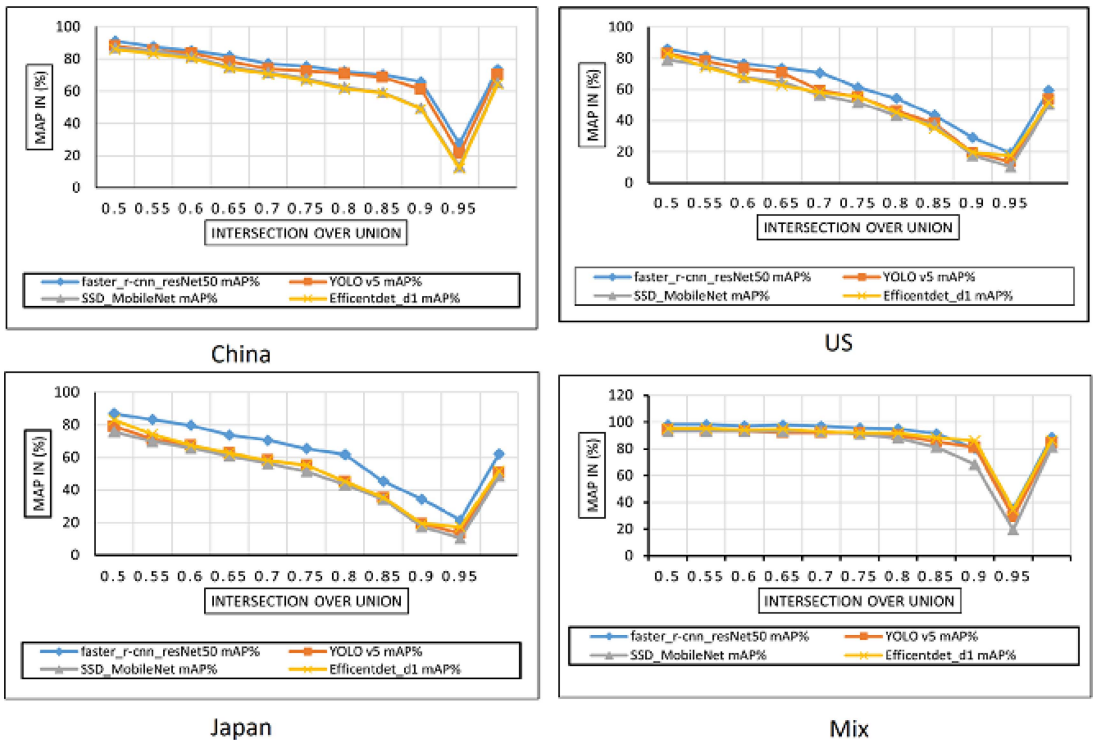


Fig. 18. Graphical representation of mAP values for trained models at all sub-data sets.

TABLE VI
COMPARISON OF RESULTS WITH STATE OF THE ART [33]

Model Name	Dataset	Class_Focused	mAP (%)
Efficientdet d0	Japan	D44	76.07
Efficientdet d0	Spain	D44	30.69
Efficientdet d0	Mix	D44	91.74
Proposed Efficientdet d1	Japan	D00, D10, D20, D40, Repair	84.18
	China	D00, D10, D20, D40, Repair	86.46
	US	D00, D10, D20, D40, Repair	81.15
	Mix	D00, D10, D20, D40, Repair	95.12

addition, it was observed that the augmentation of D40 and Repair classes instances had a positive effect on the model performance. This could be attributed to the fact that by augmenting the data set, the model had more data to learn from and thus improved its accuracy in detecting potholes and repair sections. It is also worth noting that the choice of model can

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depend on various factors, including the size and complexity of the data set and the computational resources available. Therefore, researchers and practitioners should consider the tradeoffs between model accuracy, speed, and efficiency before selecting a model for RDD and IE.

Overall, the results of the experiments indicate that OD models, such as Faster R-CNN ResNet101, YOLO5, SSD MobileNet V1 FPN, and Efficientdet D1, can be effective in accurately detecting RDs using the RDD2022 data set. By considering the strengths and limitations of each model, a suitable model can be selected for specific needs and applications.

V. Conclusion

This research work addresses the significant issue of RDD and IE by using UAVs, which has been relatively overlooked in the existing literature due to the complex nature of RDs. This study employs four distinct two-stage and one-stage detection models: Faster R-CNN, YOLOV5, SSD mobileNet V1, and Efficientdet d1, for detecting and monitoring RDs, focusing on utilizing the concept of TinyML using the RDD2022 data set. The results demonstrate that: Efficientdet d1 proves to be the most effective model, achieving the highest mean mAP values of 95.12% for focused classes in mixed category sub-data set. This is mainly attributed due to the BiFPN of the Efficientdet d1 model, which enables the detection of objects at multiple scales. Furthermore, the study findings provide insights into the significance of addressing RDD and IE and highlight that Efficientdet d1 outperforms other models in terms of efficiency by rationally scaling up the network's depth, width, and resolution. This study contributes to bridging the existing gap in RDD research, highlighting the prospects for enhancing autonomous driving systems for UAVs and setting the groundwork for the development of more reliable UAVs.

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